

No calculators are allowed on any part of this progress check. All work must be provided to receive credit. Any work completed on scratch paper will not be graded. All answers must be simplified.

1. Vector $\vec{f} = a\vec{i} - 4\vec{j}$ and $\vec{g} = \langle 2a, 8 \rangle$. If $\vec{f}$ and $\vec{g}$ are orthogonal, find all possible values of a . (2 points)

2. For the following questions, let $A = \begin{bmatrix} -1 & h \\ -2 & 1 \\ -3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} k & -2 \\ 0 & 3 \\ 5 & 1 \end{bmatrix}$

- a) If $3A + 2B = \begin{bmatrix} -1 & 5 \\ 2 & -7 \\ 16 & 5 \end{bmatrix}$, find the values of h and k . (2 points)

- b) If $C = A^T \cdot B$, find $C_{1,2}$. Show the work that leads to your answer. (2 points)

3. Solve for the matrix X (4 points)

$$X \cdot \begin{bmatrix} 7 & 3 & 3 \\ -1 & -1 & -2 \\ -6 & -3 & -4 \end{bmatrix} = \begin{bmatrix} 16 & 7 & 7 \\ 38 & 18 & 22 \end{bmatrix}$$

4. Solve the system of equations shown below by representing the system with a matrix and reducing the matrix to reduced row echelon form. Show all work. (3 points)

$$\begin{aligned} 2x - 14y &= 17 \\ 21y - 8x &= -43 \end{aligned}$$

5. Triangle ABC has vertices at $A(-a, 1)$, $B(b, 6)$, and $C(6, -6)$. It is known that the length of side BC is 13 units. If $b < 6$ and triangle ABC has an area of $\frac{11}{2}$ square units, find all possible values of a . (4 points)
6. The hyperbola given by the equation $4(x - 1)^2 - (y + 2)^2 = 16$ has a vertex at point $A(a, -2)$ where $a > 0$. The hyperbola is rotated 30° counterclockwise around the origin. Find the coordinates of A' after the rotation. (3 points)

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1. Vector $\vec{f} = 2\vec{i} - 5\vec{j}$ and $\vec{g} = \langle 2a, 8 - a \rangle$. If $\vec{f}$ and $\vec{g}$ are orthogonal, find all possible values of a . (2 points)

2. For the following questions, let $A = \begin{bmatrix} -1 & h \\ -5 & 7 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} k & -5 \\ 1 & 0 \\ -3 & -4 \end{bmatrix}$

- a) If $2A - 3B = \begin{bmatrix} -1 & 5 \\ 2 & -7 \\ 16 & 5 \end{bmatrix}$, find the values of h and k . (2 points)

- b) If $C = A^T \cdot B$, find $C_{1,2}$. Show the work that leads to your answer. (2 points)

3. Solve for the matrix X (4 points):

$$X \cdot \begin{bmatrix} -9 & 4 & 10 \\ -2 & 1 & 2 \\ 8 & -4 & -9 \end{bmatrix} = \begin{bmatrix} 25 & -13 & -28 \\ -72 & 33 & 80 \end{bmatrix}$$

4. Solve the system of equations shown below by representing the system with a matrix and reducing the matrix to reduced row echelon form. Show all work. (3 points)

$$8x + 12y = 28$$

$$3y - 6x = -13$$

5. Triangle ABC has vertices at $A(-a, -2)$, $B(b, 1)$, and $C(6, -7)$. It is known that the length of side BC is 10 units. If $b < 6$ and triangle ABC has an area of 3 square units, find all possible values of a . (4 points)
6. The hyperbola given by the equation $4(x - 1)^2 - (y + 2)^2 = 16$ has a vertex at point $A(a, -2)$ where $a < 0$. The hyperbola is rotated 60° counterclockwise around the origin. Find the coordinates of A' after the rotation. (3 points)

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1. Vector $\vec{f} = a\vec{i} - 4\vec{j}$ and $\vec{g} = \langle 2a, -8 \rangle$. If $\vec{f}$ and $\vec{g}$ are orthogonal, find all possible values of a . (2 points)

2. For the following questions, let $A = \begin{bmatrix} -h & -1 \\ -2 & 1 \\ -3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} k & -2 \\ 0 & 3 \\ 5 & 1 \end{bmatrix}$

- a) If $3A - 2B = \begin{bmatrix} -5 & 1 \\ 0 & -3 \\ 9 & -2 \end{bmatrix}$, find the values of h and k . (2 points)

- b) If $C = A^T \cdot B$, find $C_{2,2}$. Show the work that leads to your answer. (2 points)

3. Solve for the matrix X (4 points)

$$\begin{bmatrix} 12 & -1 & -11 \\ -3 & 1 & 3 \\ 1 & 0 & -1 \end{bmatrix} \cdot X = \begin{bmatrix} -14 & 59 \\ 5 & -15 \\ -1 & 5 \end{bmatrix}$$

4. Solve the system of equations shown below by representing the system with a matrix and reducing the matrix to reduced row echelon form. Show all work. (3 points)

$$\begin{aligned} 3x - 5y &= 5 \\ 10y - 15x &= -13 \end{aligned}$$

5. Triangle ABC has vertices at $A(a, -2a)$, $B(b, 4)$, and $C(2, 5)$. It is known that the length of side BC is $\sqrt{17}$ units. If $b > 2$ and triangle ABC has an area of $\frac{29}{2}$ square units, find all possible values of a . (4 points)
6. The hyperbola given by the equation $\frac{(y+2)^2}{2} - (x+3)^2 = 8$ has a vertex at point $A(-3, a)$ where $a > 0$. The hyperbola is rotated 30° clockwise around the origin. Find the coordinates of A' after the rotation. (3 points)

